STAT 578: Spectral Methods, Network Analysis, and Nonconvex Optimization

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Lecture 6
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1 Matrix Concentration Inequalities (continued)

1.1 Lieb's inequality recap

The mapping

$$A \;\mapsto\; \operatorname{Tr}\,\exp(H + \log A)$$

is concave for $A \succ 0$. This yields:

$$\mathbb{E}\,\operatorname{Tr}[\exp(H+X)] \;\le\; \operatorname{Tr}\,\exp\big(H + \log \mathbb{E}e^{X}\big).$$

1.2 Tail bound framework (Tropp style)

For self-adjoint X,

$$\Pr[\lambda_{\max}(X) \ge t] = \Pr[e^{\theta\lambda_{\max}(X)} \ge e^{\theta t}] \le e^{-\theta t}\,\mathbb{E}\,e^{\theta\lambda_{\max}(X)}.$$

Bounding $\mathbb{E}e^{\theta\lambda_{\max}(X)}$ via Lieb's inequality gives the master tail framework:

$$\Pr[\lambda_{\max}(X) \ge t] \le \inf_{\theta>0}\; e^{-\theta t}\operatorname{Tr}\,\exp\Big(\sum_k \log \mathbb{E}e^{\theta X_k}\Big).$$

2 Bernstein Inequalities

2.1 Scalar Bernstein inequality

Suppose $X_1, \dots, X_n$ are independent, mean-zero, and satisfy:

$$\mathbb{E}X_i = 0, \qquad |X_i| \le M \;\text{ a.s.}$$

Define variance proxy

$$\sigma^2 = \sum_{i=1}^n \mathbb{E}X_i^2.$$

Then for all $t > 0$,

$$\Pr\Big[|\sum_{i=1}^n X_i| \ge t\Big] \;\le\; 2\exp\left(-\frac{t^2/2}{n\sigma^2 + Mt/3}\right).$$

Special cases: Hoeffding: $X_i \in [a,b]$, then $M = b-a$, $\sigma^2 \le n(b-a)^2/4$. Bernoulli(p): $\sigma^2 = np(1-p)$, $M = 1$. Sub-Gaussian tails.

2.2 Matrix Bernstein inequality

Assume $\mathbb{E}X = 0$ and X is symmetric with

$$\lambda_{\max}(X) \le 1 \quad \text{a.s.}$$

Then for $\theta > 0$,

$$\mathbb{E}e^{\theta X} \;\preceq\; \exp\Big((e^{\theta} - \theta - 1)\,\mathbb{E}X^2\Big).$$

Therefore,

$$\Pr\Big[\,\lambda_{\max}\Big(\sum_k X_k\Big) > t\Big] \;\le\; \operatorname{tr}\,\exp\Big(-\theta t + (e^{\theta} - \theta - 1)\sum_k \mathbb{E}X_k^2\Big).$$

If $\sigma^2 = \Big\|\sum_k \mathbb{E}X_k^2\Big\|$, then

$$\Pr\Big[\,\lambda_{\max}\Big(\sum_k X_k\Big) > t\Big] \;\le\; n \cdot \exp\Big(-\theta t + (e^{\theta} - \theta - 1)\sigma^2\Big).$$

(Optimize over $\theta > 0$.)

Suppose $\{X_k\}$ are independent self-adjoint random matrices with

$$\mathbb{E}X_k = 0, \qquad \lambda_{\max}(X_k) \le R \quad \text{a.s.}$$

Then

$$\Pr\Big[\lambda_{\max}\Big(\sum_k X_k\Big) \ge t\Big] \;\le\; d \cdot \exp\left(-\frac{t^2/2}{\sigma^2 + Rt/3}\right),$$

where d is the matrix dimension.

This is the matrix generalization of the scalar Bernstein bound.

2.3 Gaussian tail bound

For $X_i \sim \mathcal{N}(0, \sigma^2)$,

$$\Pr\Big[\Big|\sum_{i=1}^n X_i\Big| > t\Big] \;\le\; 2\exp\left(-\frac{t^2}{2n\sigma^2}\right).$$

Hence

$$\Pr\left[\Big|\frac{1}{n}\sum_{i=1}^n X_i\Big| > \frac{s\sigma}{\sqrt{n}}\right] \;\le\; 2\exp\left(-\frac{s^2}{2}\right).$$

Choosing $s = \sqrt{100\log n}$, we get

$$\Big|\frac{1}{n}\sum_{i=1}^n X_i\Big| \;\le\; \frac{\sigma}{\sqrt{n}} \cdot \sqrt{100\log n} \;=\; \sigma\sqrt{\frac{100\log n}{n}}$$

with probability at least $1 - 2n^{-50}$.

2.4 Bai–Yin law

Suppose X is a Wigner matrix, i.e. $X_{ij} = X_{ji}$ and

$$\mathbb{E}X_{ij}^2 = \sigma^2, \qquad \mathbb{E}X_{ii}^2 = 2\sigma^2.$$

Then

$$\frac{\|X\|}{\sqrt{2\sigma^2 n}} \xrightarrow{\text{a.s.}} 1.$$

Goal. We want a non-asymptotic estimate:

$$\|X\| \;\approx\; \sqrt{2\sigma^2 n}\,\big[\,1 + o(1)\,\big].$$

Bandira and van Handel (2015, Annals of Prob.) "Sharp nonasymptotic bounds."

$$\mathbb{E}\|X\| \;\le\; \Big(\mathbb{E}\|X\|_F^k\Big)^{1/k}.$$

Remark 3.13. For $\sigma^2 = \max_j \sum \mathbb{E}X_{ij}^2$ and $\sigma_*^2 = \max_{ij} \mathbb{E}X_{ij}^2$, one has

$$\Pr\Big[\|X\| \;\ge\; (1+\varepsilon)\,2\sqrt{2}\sigma + t\Big] \;\le\; n\exp\Big(-\tfrac{t^2}{c\sigma_*^2}\Big),$$

where $c > 0$.

3 Gaussian Orthogonal Ensemble (GOE)

$$X \sim \mathrm{GOE}(n)$$

if

$$X_{ii} \sim \mathcal{N}(0, \sigma^2), \quad X_{ij} \sim \mathcal{N}(0, \tfrac{\sigma^2}{2}) \text{ for } i < j, \quad X_{ij} = X_{ji}.$$

This defines the Gaussian Orthogonal Ensemble.

If W is an orthogonal matrix, then

$$WXW^\top \;\overset{d}{=}\; X.$$

Hence, for any fixed unit vector z,

$$z^\top X z \;\sim\; \mathcal{N}(0, \|z\|^2).$$

Moreover, if $X = U\Lambda U^\top$ is the eigen-decomposition, then

$$WU \;\overset{d}{=}\; U,$$

which implies the eigenvectors of X are Haar-distributed (i.e. uniform on orthogonal matrices).

4 ε–net argument

$$\|X\| = \sup_{\substack{z,y \\ \|z\|=1,\,\|y\|=1}} z^\top X y.$$

An ε–net of the unit sphere

$$S^{d-1} = \{z \in \mathbb{R}^d : \|z\| = 1\}$$

is a finite (or countable) set $\mathcal{N} \subseteq S^{d-1}$ such that for every $z \in S^{d-1}$ there exists $z' \in \mathcal{N}$ with

$$\|z - z'\| \le \varepsilon.$$

Thus one can approximate the operator norm by restricting the supremum to $\mathcal{N}$:

$$\|X\| \approx \sup_{z',y' \in \mathcal{N}} (z')^\top X y'.$$